

Determinants

T.T.

The determinant is a single number with an unusual amount of information about a matrix. It detects invertibility, records the effect of elimination, and eventually produces formulas for inverses and solutions. Its linearity applies to one row at a time: the tempting rule $\det(A + B) = \det A + \det B$ is false.

Adapted from Gilbert Strang, Section 5.1, pp. 247–248.

DEFINITION AND PROPERTIES

$$\det I = 1, \quad \det(AB) = \det(A) \det(B), \quad A \text{ invertible} \iff \det A \neq 0.$$

Row rules:

swapping rows changes the sign; multiplying a row by c multiplies the determinant by c ; adding a multiple of one row to another leaves it unchanged.

Linearity:

the determinant is linear in each row (or column) separately and alternating in its rows.

$$\det(A^T) = \det A, \quad \det(A^{-1}) = \det A^{-1}, \quad \det(cA) = c^n \det A.$$

COMPUTATIONAL FORMULAE

Triangular matrices:

$$\det A = \prod_{i=1}^n a_{ii}.$$

$$\det A = \sum_{j=1}^n a_{ij} C_{ij} = \sum_{i=1}^n a_{ij} C_{ij}, \quad C_{ij} = (-1)^{i+j} \det(A_{ij}).$$

Cofactor formula:

$$A^{-1} = \frac{1}{\det A} \text{Cof}(A)^T.$$

$$\det A = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n a_{i, \sigma(i)}.$$

Cramer's rule:

$$\text{if } Ax = b \text{ and } \det A \neq 0, x_i = \frac{\det A_i(b)}{\det A}.$$

Vandermonde determinant:

$$\det \left([x_j^{i-1}]_{i,j=1}^n \right) = \prod_{1 \leq i < j \leq n} (x_j - x_i).$$

GEOMETRY AND VOLUME

$$\text{volume}(A([0, 1]^n)) = |\det A|, \quad \det A = 0 \iff \text{columns lie in a lower-dimensional subspace.}$$

PROBLEM SET

Q1.

Use row operations to compute

$$\det \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 1 & 1 & 1 \end{bmatrix}.$$

State the determinant rule used at each step and identify the product of the pivots.

Q2.

Let A be an $n \times n$ matrix and let $c \in \mathbb{R}$.

- (i) Use linearity in one row at a time to show that

$$\det(cA) = c^n \det A.$$

- (ii) If A is 4×4 and $\det A = -2$, find $\det(2A)$, $\det(A^T A)$, and $\det(A^{-1})$.
(iii) Show by a 2×2 counterexample that the determinant is not linear as a function of the whole matrix: in general, $\det(A + B) \neq \det A + \det B$.

Q3.

Let A be an $n \times n$ matrix and let $\mathbf{1} = (1, \dots, 1)^T$.

- (i) Suppose the entries in every row of A add to zero. Show that $A\mathbf{1} = 0$, and hence that $\det A = 0$.
(ii) Suppose instead that the entries in every row add to one. Show that $\det(A - I) = 0$.
(iii) Does the condition in part (ii) imply that $\det A = 1$? Give a counterexample and explain what the condition actually proves.

Q4.

Let $A = [a_1 \ \cdots \ a_n]$ be an $n \times n$ matrix.

- (i) Assume that the columns of A are linearly dependent. Use linearity of the determinant in one column and the repeated-column rule to prove that $\det A = 0$.
(ii) Prove the converse using the connection between the determinant and invertibility.
(iii) Explain why this equivalence agrees with the volume interpretation of $|\det A|$.

Q5.

Let Q be a real orthogonal matrix, so that $Q^T Q = I$.

- (i) Prove that $\det Q = 1$ or $\det Q = -1$.
(ii) For a 2×2 orthogonal matrix, write its first column as $(\cos \theta, \sin \theta)^T$. Show that its second column must be either $(-\sin \theta, \cos \theta)^T$ or $(\sin \theta, -\cos \theta)^T$, and identify the determinant in each case.

Q6.

Suppose that A is an invertible matrix with integer entries.

- (i) Prove that if $\det A = 1$ or $\det A = -1$, then every entry of A^{-1} is an integer.
- (ii) Prove the converse: if both A and A^{-1} have integer entries, then $\det A = 1$ or $\det A = -1$.

Q7.

Let $A = [a_1 \ \cdots \ a_n]$ be invertible, and let $A_j(b)$ denote the matrix obtained by replacing column j of A by b . Suppose that $Ax = b$.

- (i) Write b as a linear combination of the columns of A .
- (ii) Use linearity of the determinant in column j to show that

$$\det A_j(b) = x_j \det A.$$

Explain why every other term in the expansion is zero.

- (iii) Deduce Cramer's rule.

Q8.

Let $C = (C_{ij})$ be the cofactor matrix of an $n \times n$ matrix A .

- (i) Show that every diagonal entry of AC^T equals $\det A$.
- (ii) Show that every off-diagonal entry of AC^T is zero. Hint: interpret it as a cofactor expansion of a matrix with two equal rows.
- (iii) Deduce the identity

$$AC^T = (\det A)I,$$

and hence derive the cofactor formula for A^{-1} when $\det A \neq 0$.